

Understanding the representation of climate impacts and adaptation in Integrated Assessment Models



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Motivation

- IAMs excel at representing mitigation pathways and technological transitions but typically have **less advanced treatment of climate impacts and adaptation**.
- **Fragmentation** across model types (sectoral models, cost-benefit IAMs, process-based IAMs) leads to systematic **misrepresentation** of climate damage timing, magnitude, and distribution.
- **Adaptation** is critically under-represented.
- Particularly problematic for assessing regional vulnerabilities and **mitigation-adaptation interactions**.

This paper: We run a novel systematic **global survey of climate impacts and adaptation representation in IAMs**, designed to capture both quantitative and qualitative aspects of impact and adaptation representation across models.

Global survey of IAMs

- Survey sent to >100 IAM teams identified through IAMC wiki, IPCC AR6 contributing models, and expert recommendations.
- Designed to capture both quantitative and qualitative aspects of impact and adaptation representation.
- 14 dimensions of impacts and adaptation

For each impact category, respondents indicated:

- Impact representation (not represented, 1-directional, 2-directional, planned)
- Inclusion in optimization routines
- Modeling approach (monetary, biophysical, or both)
- Main outcome variables and units
- Adaptation representation and mechanisms
- Adaptation optimization (optimal, partial fixed, variable non-optimal)

Responses

>40 teams participating

Model type	%
Process-Based IAMs	32.5
CGE	25.0
Energy System Optimization	17.5
Cost-Benefit Analysis	7.5
Agent-Based and Behavioral	5.0
Hybrid/Multi-Method	5.0
Partial Equilibrium	5.0
System Dynamics	2.5

Contribute to the survey

Scan the QR code or contact us.
One contributor per team **will be co-author**.



Main results

The Integration Gap. IAMs are still primarily energy-economic models with limited sectoral breadth rather than truly integrated assessment tools

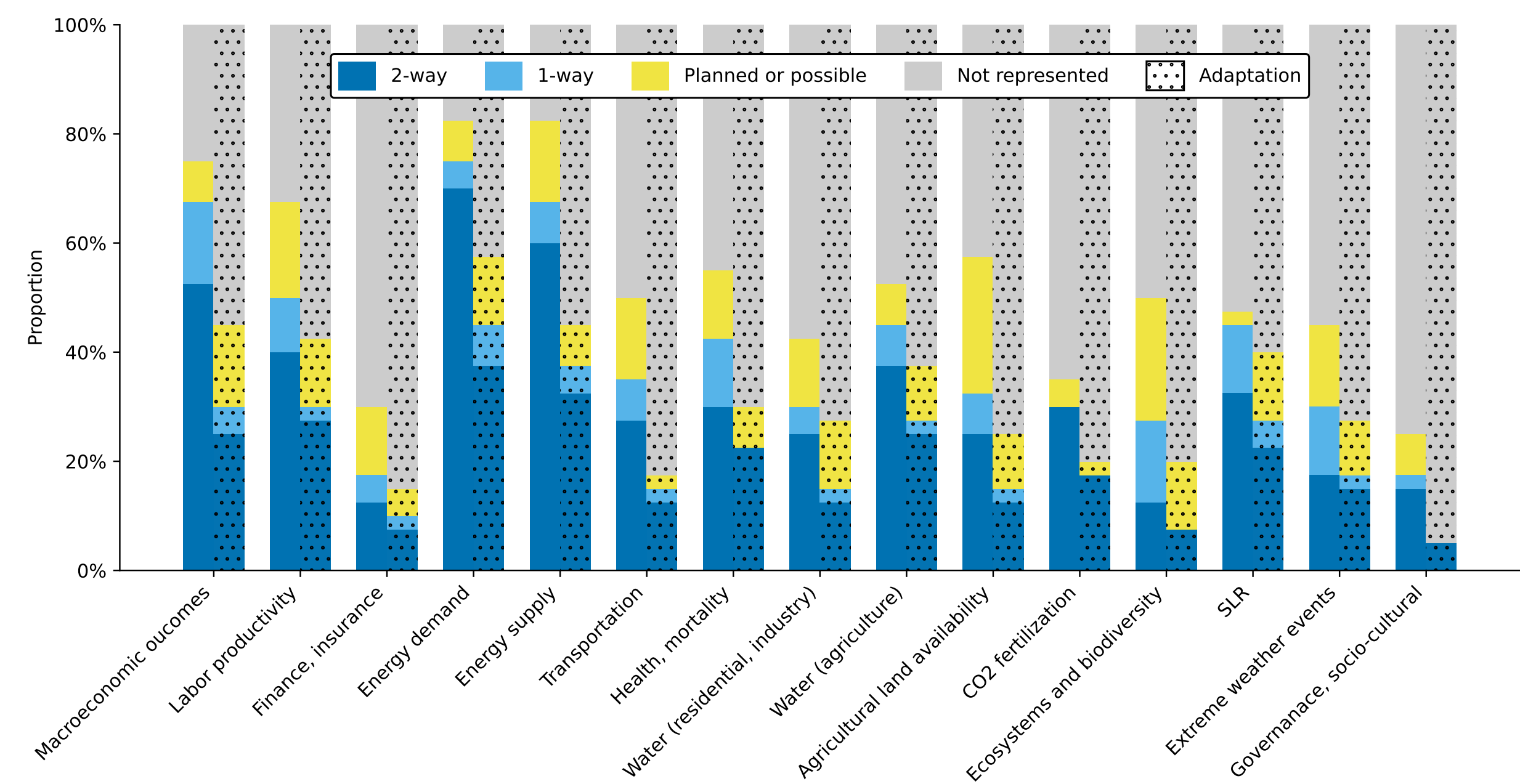
The Adaptation Gap. Inclusion of adaptation in models lags behind in most dimensions, especially in ecosystem services and biodiversity, and agricultural land availability.

The Monetary-Biophysical divide. Economic dimensions favor monetary approaches while non-economic dimensions more often use biophysical measures, creating integration barriers for policy-relevant trade-offs.

The Optimization Paradox. Very few models optimize adaptation, yet this is exactly what policymakers need for efficient adaptation planning.

The Policy Relevance Gap. The concentration in energy-economic sectors misaligns with NDC priorities and Global Stocktake needs, which require broader sectoral coverage.

Figure 1. Share of models representing impacts and adaptation. Impacts on energy and the economy are the most represented. Adaptation lags behind across all dimensions.



Surveyed models cluster in 3 groups:
(1) models mostly representing economic impacts and sea level rise;
(2) models mostly representing energy systems;
(3) comprehensive models representing multiple dimensions.

	1	2	3	N
Process-Based IAMs	36%	29%	33%	13
CGE Models	29%	0%	50%	10
Energy System Optimization	7%	43%	0%	7
Cost-Benefit Analysis	21%	0%	0%	3
Agent-Based and Behavioral	7%	7%	0%	2
Hybrid/Multi-Method	0%	0%	17%	2
Partial Equilibrium Models	0%	14%	0%	2
System Dynamics Models	0%	7%	0%	1

Models clustered using Kmodes.
Models: 7see, AIM, AIM-Technology, AIM/Hub-Korea, AIM/Hub-Vietnam, BLUES 2.0 C3IAM, CCEM, COIN, DSK, DYNERIO, E4SMA-EU-TIMES 1.0, ENV-Linkages, ENVISAGE, FRIDA2.1, GCAM-KSA, GEM-E3, GEMINI-E3, GENeSYS-MOD, GRACE, ICES, ICES-EMF, IF, IMACLIM-R, IMAGE 3.4, MagPIE, MEDEAS-V, MESSAGEix-GLOBIOM, MIMOSA, MUSE, NEMESIS, NICE2020, POLES, PROMETHEUS, REMIND, REMod, RICE50+, TIMES-Sweden, TIMES-PT, WITCH.

